Bidirectional lenses, synthesis, and categorical framing

Bidirectional transformations and the lens view of COGANT

The bidirectional transformation literature offers the cleanest mathematical characterization of what COGANT does at the architectural level. A *lens* in the sense of Foster et al. [@foster2007lenses] is a pair of functions $\texttt{get} : S \to A$ and $\texttt{put} : A \to S \to S$ satisfying two round-trip laws: $\texttt{get}(\texttt{put}(a, s)) = a$ (PutGet) and $\texttt{put}(\texttt{get}(s), s) = s$ (GetPut). COGANT's `cogant.translate` (forward) and `cogant.reverse` (reverse) modules are precisely such a pair, with S the AST-level program graph of a repository and A the emitted Generalized Notation Notation bundle. Neither law holds exactly in v0.5.x — the confidence model introduces lossy abstraction — but the lens framing makes the deviation measurable: the confidence tier of each assertion is a quantitative record of how far the extraction falls short of an exact `get`, and the round-trip isomorphism score of §9 and `../cogant/docs/evaluation/ISOMORPHISM_THEOREM.md` is an empirical estimate of how far a recovered program differs from its